

Equation of motion

$$x[k+1] = x[k] + a_x[k] \{f_{dx}[k] + f_T[k]\} + \delta x_D[k], \quad a_x = \Delta t / \gamma(h)$$

Measurement

(d -step delay)

$$\delta x_m[k] = \delta x[k-d] + n_x[k], \quad \delta x = x_d - x$$

Control effort

$$f_{dx}[k] = \hat{a}_x^{-1}[k] \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x_m[k] - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}$$

Motion error

$$\delta x[k+1] = \lambda_c \delta x[k] - F_{dx}[k] e_{a_x}[k] - \underbrace{\left\{ \delta x_T^d[k] - \delta x_{\delta af}^d[k] + (1 - \lambda_c) n_x[k] \right\}}_{\varepsilon[k]}$$

$$F_{dx}[k] = f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k-i], \quad e_{a_x} = a_x - \hat{a}_x, \quad \sigma_{\delta h}^2 := 4k_B T a_x$$

$$\sigma_\varepsilon^2 := \gamma_\varepsilon(0) = (1+2(1-\lambda_c)^2) \sigma_{\delta h}^2 (= Q_{33}); \quad \text{AR(1) white driver: } (3-2\lambda_c^2) \sigma_{\delta h}^2; \quad \frac{3-2\lambda_c^2}{1+2(1-\lambda_c)^2} = 1.712 \ (\lambda_c=0.7)$$

Gain model (first-order Taylor about the desired trajectory)

$$x[k+1] = x_d[k+1] - \delta x[k+1]$$

$$a_x[k+1] = a_x(x[k+1]) = \underbrace{a_x(x_d[k+1])}_{a_{x_{det}}[k+1]} - a'_x[k+1] \delta x[k+1]$$

$$a_{x_{det}}[k+1] = a_{x_{det}}[k] + a'_x[k] \Delta h_d[k], \quad \Delta h_d[k] = h_d[k+1] - h_d[k], \quad a'_x[k+1] \approx a'_x[k] =: a'_x$$

True recursion

$$(a_{x_{det}}[k] = a_x[k] + a'_x \delta x[k], \text{ then the boxed } \delta x \text{ recursion})$$

$$a_x[k+1] = a_x[k] + a'_x \Delta h_d[k] + (1 - \lambda_c) a'_x \delta x[k] + a'_x F_{dx}[k] e_{a_x}[k] + a'_x \varepsilon[k]$$

Estimator (predict + update)

$$(E\{e_{a_x}\} = 0, E\{\varepsilon\} = 0; \delta x[k] \rightarrow \delta \hat{x}_3[k])$$

$$\hat{a}_x[k+1] = \hat{a}_x[k] + a'_x \Delta h_d[k] + (1 - \lambda_c) a'_x \delta \hat{x}_3[k] + \ell_{41}[k] e_{y_1}[k] + \ell_{42}[k] e_{y_2}[k]$$

Error dynamics

$$(e_{\delta x_3} = \delta x_3 - \delta \hat{x}_3)$$

$$e_{a_x}[k+1] = (1 + a'_x F_{dx}[k]) e_{a_x}[k] + (1 - \lambda_c) a'_x e_{\delta x_3}[k] + \underbrace{a'_x \varepsilon[k]}_{q_4[k]} - \ell_{41}[k] e_{y_1}[k] - \ell_{42}[k] e_{y_2}[k]$$

$\mathbf{F}_e$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & (1 - \lambda_c) a'_x & 1 + a'_x F_{dx}[k] \end{bmatrix}, \quad \text{eig}(\mathbf{F}_e) = \{0, 0, \lambda_c + a'_x F_{dx}, 1\}$$

Process noise (rank-1)

$$(q_3 = -\varepsilon, q_4 = a'_x \varepsilon: \text{one sample})$$

$$\mathbf{q}[k] = \varepsilon[k] \begin{bmatrix} 0 \\ 0 \\ -1 \\ a'_x \end{bmatrix} \Rightarrow \mathbf{Q}[k] = \sigma_\varepsilon^2 \begin{bmatrix} 0 \\ 0 \\ -1 \\ a'_x \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -1 \\ a'_x \end{bmatrix}^\top$$

$$Q_{33} = \sigma_\varepsilon^2, \quad Q_{34} = Q_{43} = -a'_x \sigma_\varepsilon^2, \quad Q_{44} = a'^2_x \sigma_\varepsilon^2$$

$$P_{44} = a'^2_x P_{33}, \quad P_{34} = -a'_x P_{33} \quad (\text{exact filter invariant, } \forall R_{22})$$

$$\frac{\gamma_\varepsilon(0)}{1-\lambda_c^2} = 2.31 \sigma_{\delta h}^2 \text{ vs } \text{Var}(\delta x) = C_{\delta x} \sigma_{\delta h}^2 = 3.96 \sigma_{\delta h}^2, \quad C_{\delta x} = 2 + \frac{1}{1-\lambda_c^2} \quad (\varepsilon \text{ is MA}(d), \text{ not white})$$

Feedforward (implemented forms)

$$\text{known: } \Delta \hat{a} = a_{x_{det}}[k] - a_{x_{det}}[k-1]; \quad \text{ahat: } \Delta \hat{a} = \hat{a}_x \left(\frac{a_{x_{det}}[k]}{a_{x_{det}}[k-1]} - 1 \right); \quad \text{diff: } \Delta \hat{a} = \hat{a}'_x \Delta h_d[k-1]$$

a'_x acquisition

$$\text{known: } a'_x = -\frac{a_{x_{det}}(\bar{h}_d) K_h(\bar{h}_d)}{R}; \quad \text{ahat: } a'_x = -\frac{\hat{a}_x[k] K_h(\bar{h}_d)}{R}; \quad K_h = \frac{1}{c} \frac{dc}{dh}$$

$$\text{diff: } a'_{raw}[k] = \frac{\hat{a}_x[k] - \hat{a}_x[k-1]}{\Delta h_d[k-1]}, \quad \text{update iff } |\Delta h_d[k-1]| \geq \gamma$$

$$\hat{a}'_x[k] = (1 - \beta) \hat{a}'_x[k-1] + \beta a'_{raw}[k], \quad \beta = 0.05, \quad \gamma = 10^{-3} \mu\text{m}; \quad \hat{a}'_x \leftarrow \max(\hat{a}'_x, 0) \text{ (optional)}$$

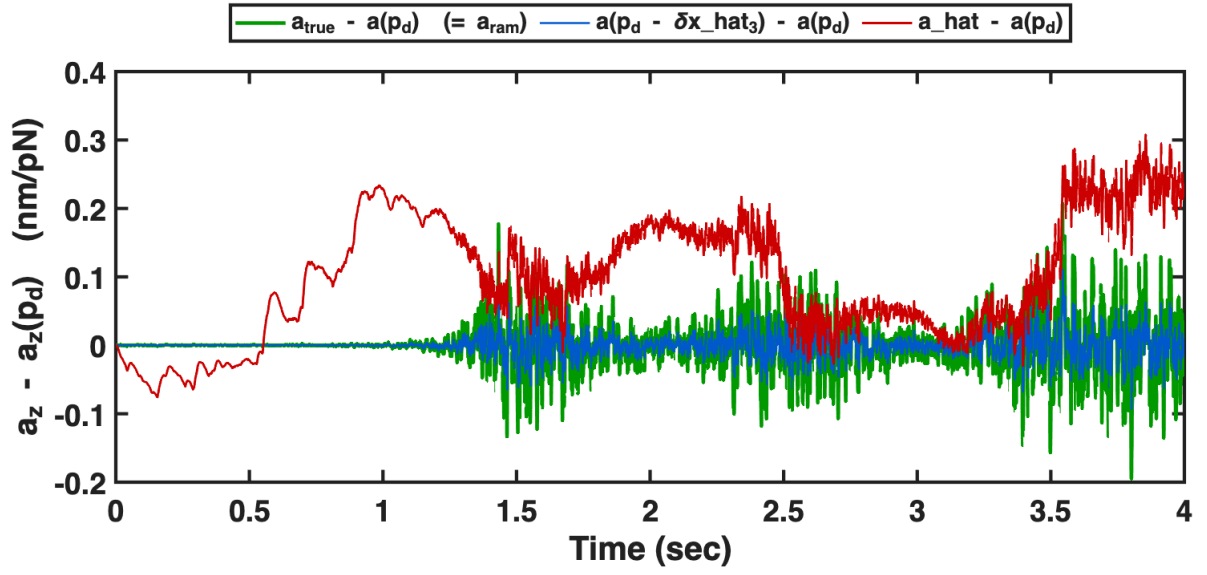
R_{22} delay term

$$(r_2 = n_a - \sum_{i=1}^d \Delta a_{ram}[k-i], d=2)$$

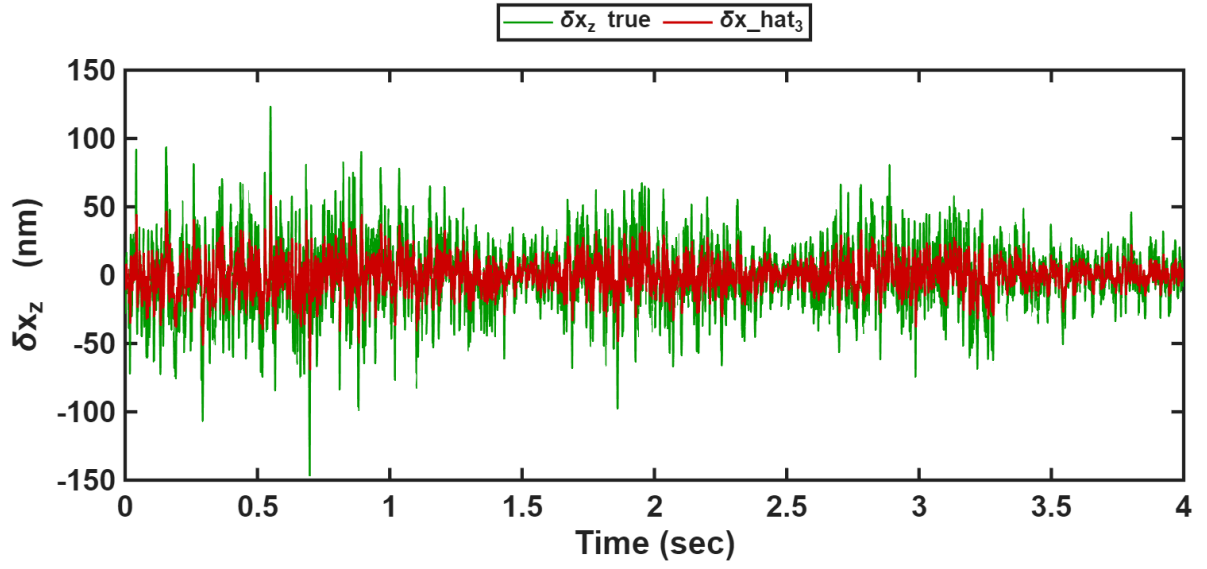
$$\Delta a_{ram}[k-1] + \Delta a_{ram}[k-2] = a_{ram}[k] - a_{ram}[k-2] \Rightarrow \text{Var}\left(\sum\right) = 2(1 - \rho_2) \text{Var}(a_{ram})$$

$$R_{22}^{delay} = \frac{1 - \rho_2}{2(1 - \rho_1)} \sum_{i=1}^d \text{Var}(\Delta a_{ram}[k - i]), \quad \frac{1 - \rho_2}{2(1 - \rho_1)} = 1.105 \ (\lambda_c=0.7)$$

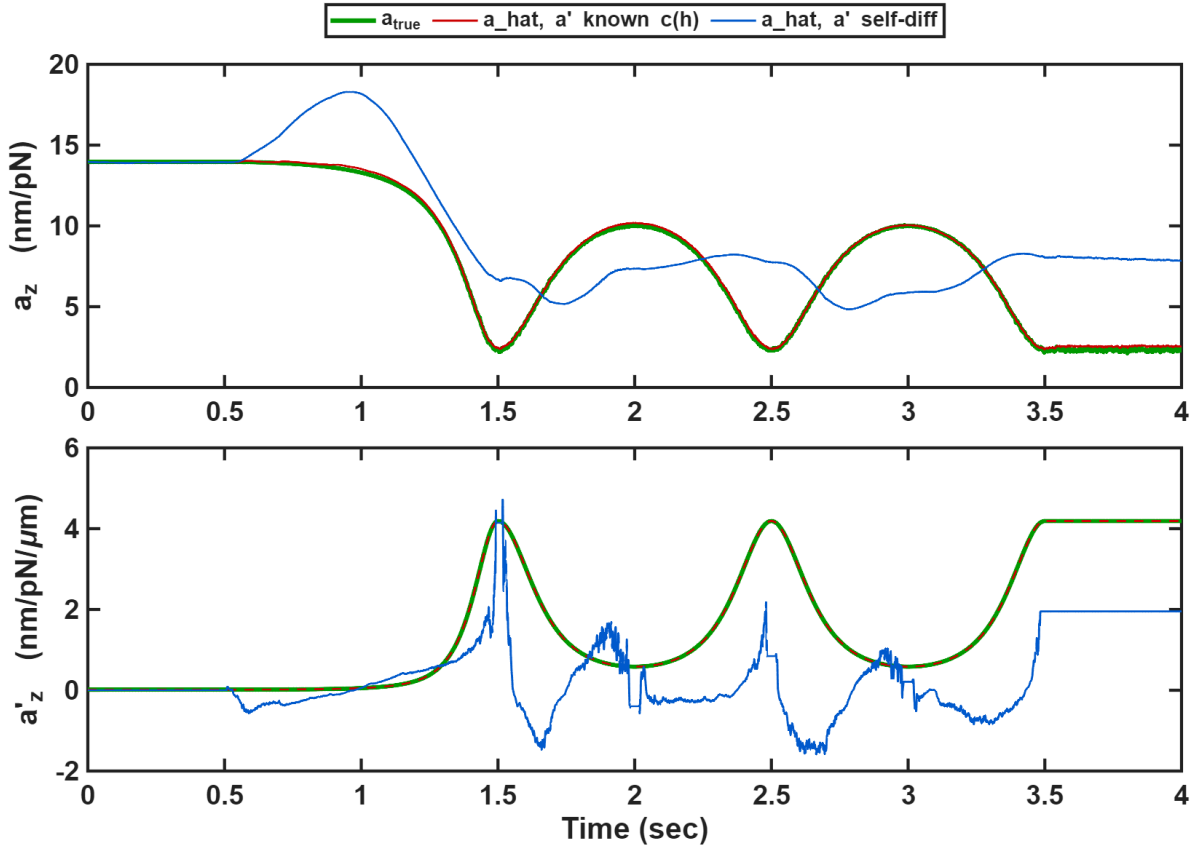
$$\rho_1 = 0.8515, \quad \rho_2 = 0.6718 \quad (\rho_\tau = \text{lag-}\tau \text{ autocorr of } \delta x)$$



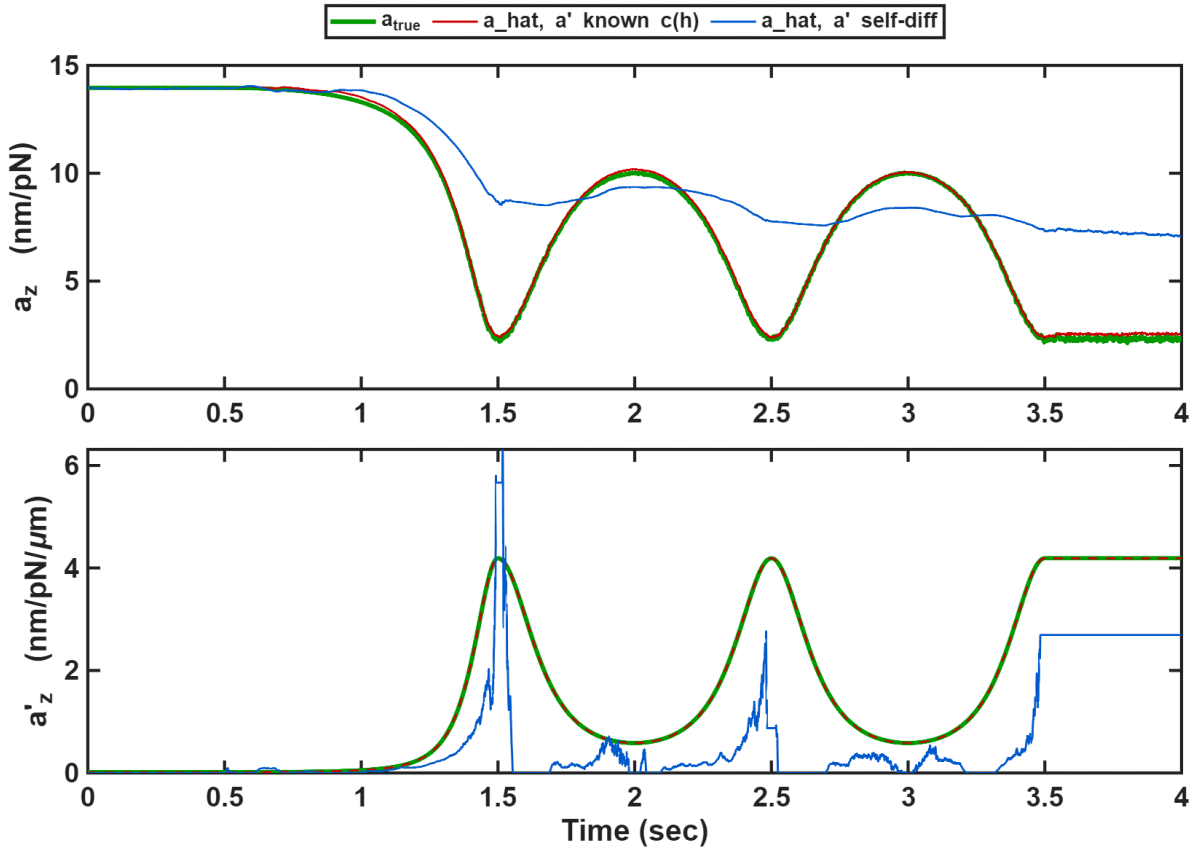
Oracle- a' arm, z axis, 1 Hz osc, seed 1: $\hat{a}_x - a_x(p_d)$ vs $a_{x,\text{true}} - a_x(p_d) (= a_{x_{\text{ram}}})$.



Same arm and seed: $\delta \hat{x}_3$ vs true δx_z .



Same seed: $\hat{a}_x$ (top) and a'_x (bottom), oracle- a' arm vs self-diff arm (no clamp).



Same seed with the post-EWMA clamp $\hat{a}'_x \leftarrow \max(\hat{a}'_x, 0)$.